



CloudHustler Commerce Platform Troubleshooting Guide

This document captures the major issues encountered during the development of the **CloudHustler Commerce Platform**, along with their root causes, troubleshooting process, and final resolution.

The goal is to provide an operational runbook for future deployments while documenting the engineering decisions and lessons learned throughout the project.

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Bootstrap

Issue #1 – Terraform Backend Initialization

Symptoms

- Backend initialization failures

- Remote state configuration errors

Root Cause

Terraform backend resources did not yet exist.

Resolution

- Created bootstrap infrastructure locally.
- Provisioned:
 - KMS Key
 - S3 Backend Bucket
 - DynamoDB Lock Table
- Migrated local state to the remote backend.

Lesson Learned

Terraform backends cannot manage themselves. Always bootstrap infrastructure separately before enabling remote state.

Issue #2 – S3 Bucket Versioning and Deletion

Symptoms

BucketNotEmpty

Root Cause

Bucket versioning was enabled.

Deleting current objects does not remove previous object versions.

Resolution

Removed all object versions and delete markers before deleting the bucket.

Lesson Learned

Version-enabled buckets require deleting **every version** before the bucket can be removed.

Issue #3 – Remote State Migration

Symptoms

Terraform reported backend configuration changes.

Resolution

Migrated state using:

```
terraform init -migrate-state
```

Lesson Learned

Always migrate existing state rather than recreating infrastructure.

Organization

Issue #4 – AWS Organizations Deployment

Symptoms

```
AlreadyInOrganizationException
```

Root Cause

The AWS account already belonged to an existing organization.

Resolution

Imported or reused the existing organization instead of attempting to create another one.

Lesson Learned

Always verify existing AWS Organizations before provisioning new ones.

Identity

Issue #5 – IAM Policy Generation

Symptoms

Terraform validation failed when generating IAM policies.

Root Cause

Improper JSON formatting.

Resolution

Generated IAM policies using `jsonencode()` instead of manually writing JSON.

Lesson Learned

Always use Terraform-native functions to generate IAM policy documents.

Issue #6 – IAM Module Refactoring

Symptoms

Limited flexibility for attaching managed policies.

Resolution

Refactored the IAM module to support:

- Custom policy documents
- AWS-managed policies
- Customer-managed policies
- Permission boundaries
- Instance profiles

Lesson Learned

Design reusable modules around multiple production use cases.

Governance

Issue #7 – CloudTrail CloudWatch Logs Integration

Symptoms

```
InvalidCloudWatchLogsLogGroupArnException
```

Root Cause

CloudTrail required both:

- CloudWatch Log Group
- IAM Role

Resolution

Created reusable CloudWatch Log Group and CloudTrail service role modules.

Issue #8 – AWS Config Delivery Channel

Symptoms

```
InsufficientDeliveryPolicyException
```

Root Cause

AWS Config lacked permission to write to the S3 bucket.

Resolution

Added the required bucket policy.

Issue #9 – Access Analyzer Naming Requirements

Symptoms

Terraform apply failed.

Root Cause

Analyzer names must begin with a letter.

Resolution

Updated naming convention.

Issue #10 – Config Aggregator Organization Role

Symptoms

Organization aggregation failed.

Root Cause

Missing IAM role.

Resolution

Created a dedicated IAM role for the Config Aggregator.

Issue #11 – KMS Permissions for CloudWatch Logs

Symptoms

CloudWatch failed to encrypt logs.

Root Cause

Missing KMS permissions.

Resolution

Updated the KMS key policy to allow CloudWatch Logs service access.

Networking

Issue #12 – Security Group Rule Processing

Symptoms

Terraform validation errors involving security group rules.

Root Cause

Improper precedence between:

- Security Group references
- IPv4 CIDRs
- IPv6 CIDRs
- Prefix Lists

Resolution

Refactored rule processing using explicit precedence logic.

Lesson Learned

Only one destination type should be configured for a single rule.

Issue #13 – ALB Listener Configuration

Symptoms

```
Too few blocks specified for "default_action"
```

Root Cause

Listener configuration did not follow the AWS provider schema.

Resolution

Rewrote listener resources to support:

- Forward
- Redirect
- Fixed Response

using the native AWS action model.

Lesson Learned

Follow provider schemas exactly rather than simplifying resource definitions.

Issue #14 – ACM Hosted Zone Configuration

Symptoms

DNS validation records were not created correctly.

Root Cause

Hosted zone lookup ambiguity.

Resolution

Refactored the ACM module to accept:

```
hosted_zone_id
```

instead of performing automatic lookups.

Lesson Learned

Passing the hosted zone ID explicitly makes reusable modules more predictable.

Issue #15 – Route 53 DNS Validation

Symptoms

ACM validation records appeared correct but certificate validation did not complete.

Root Cause

DNS validation depends on public DNS resolution.

Resolution

Verified:

- Hosted Zone
- Validation Records
- DNS propagation

- Public DNS visibility
-

Issue #16 – ACM Certificate Stuck in PENDING_VALIDATION

Symptoms

Terraform timed out waiting for ACM:

```
waiting for ACM Certificate to be issued  
  
PENDING_VALIDATION
```

ALB creation also failed:

```
UnsupportedCertificate
```

Investigation

Verified:

- Correct domain
- Correct hosted zone
- Correct validation CNAME records
- Correct Route 53 configuration

Everything inside AWS was configured correctly.

Root Cause

The domain registrar was **not delegating the domain to the Route 53 authoritative name servers**.

Although the validation records existed inside Route 53, ACM validates using **public DNS**, so it could not see the records.

Resolution

Updated the registrar to use the Route 53 name servers:

```
ns-896.awsdns-48.net.  
ns-1328.awsdns-38.org.
```

```
ns-1630.awsdns-11.co.uk.  
ns-469.awsdns-58.com.
```

After DNS propagation:

```
aws acm describe-certificate  
  --certificate-arn <certificate-arn>  
  --query "Certificate.Status"
```

returned:

```
ISSUED
```

Re-running:

```
terraform apply
```

completed successfully.

Lessons Learned

Creating a Route 53 hosted zone **does not automatically update the domain registrar**.

The registrar must delegate the domain to the Route 53 name servers before ACM can validate ownership.

Best Practices

Throughout this project, several engineering principles have proven valuable:

- Bootstrap shared infrastructure before enabling remote state.
 - Design reusable modules that support production use cases.
 - Use `jsonencode()` for IAM policies.
 - Prefer explicit configuration over implicit resource discovery.
 - Validate infrastructure with `terraform validate` and `terraform plan` before applying.
 - Follow AWS provider schemas exactly.
 - Verify DNS delegation whenever working with ACM or Route 53.
 - Build modules incrementally and ensure each resource is production-ready before moving to the next component.
 - Document every significant issue and its resolution to create an operational knowledge base.
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Future Entries

This document should continue to grow as additional phases are completed, including:

- Security
- Amazon ECR
- Amazon EKS
- GitOps
- Argo CD
- Helm
- Kubernetes Networking
- Observability
- CI/CD
- Disaster Recovery
- Multi-Region Deployments
- Cost Optimization
- AI Operations

Over time, this document will serve as the operational runbook and engineering knowledge base for the CloudHustler Commerce Platform.